

Randomized, Double-Blind, Placebo-Controlled Phase II Study of AMG 386 Combined With Weekly Paclitaxel in Patients With Recurrent Ovarian Cancer

Beth Y. Karlan, Amit M. Oza, Gary E. Richardson, Diane M. Provencher, Vincent L. Hansen, Martin Buck, Setsuko K. Chambers, Prafull Ghatage, Charles H. Pippitt Jr, John V. Brown III, Allan Covens, Raj V. Nagarkar, Margaret Davy, Charles A. Leath III, Hoa Nguyen, Daniel E. Stepan, David M. Weinreich, Marjan Tassoudji, Yu-Nien Sun, and Ignace B. Vergote

See accompanying article on page 441 and editorial on page 345; listen to the podcast by Dr Konstantinopoulos at www.jco.org/podcasts

Author affiliations appear at the end of this article.

Submitted December 23, 2010; accepted September 16, 2011; published online ahead of print at www.jco.org on December 19, 2011.

Supported by Amgen.

Presented in part at the 46th Annual Meeting of the American Society of Clinical Oncology, June 4-8, 2010, Chicago, IL, and the 35th European Society of Medical Oncology Annual Congress, October 8-12, 2010, Milan, Italy.

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Authors' disclosures of potential conflicts of interest and author contributions are found at the end of this article.

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Corresponding author: Beth Y. Karlan, MD, Division of Gynecologic Oncology, Cedars-Sinai Medical Center, Los Angeles, CA, 90048; e-mail: Beth.Karlan@cshs.org.

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0732-183X/12/3004-362/\$20.00

DOI: 10.1200/JCO.2010.34.3178

ABSTRACT

Purpose

To estimate the efficacy and toxicity of AMG 386, an investigational peptide-Fc fusion protein that neutralizes the interaction between the Tie2 receptor and angiopoietin-1/2, plus weekly paclitaxel in patients with recurrent ovarian cancer.

Patients and Methods

Patients with recurrent epithelial ovarian, fallopian tube, or primary peritoneal cancer were randomly assigned 1:1:1 to receive paclitaxel (80 mg/m² once weekly [QW], 3 weeks on/1 week off) plus intravenous AMG 386 10 mg/kg QW (arm A), AMG 386 3 mg/kg QW (arm B), or placebo QW (arm C). The primary end point was progression-free survival (PFS). Secondary end points included overall survival, objective response, CA-125 response, safety, and pharmacokinetics.

Results

One hundred sixty-one patients were randomly assigned. Median PFS was 7.2 months (95% CI, 5.3 to 8.1 months) in arm A, 5.7 months (95% CI, 4.6 to 8.0 months) in arm B, and 4.6 months (95% CI, 1.9 to 6.7 months) in arm C. The hazard ratio for arms A and B combined versus arm C was 0.76 (95% CI, 0.52 to 1.12; *P* = .165). Further analyses suggested an exploratory dose-response effect for PFS across arms (Tarone's test, *P* = .037). Objective response rates for arms A, B, and C were 37%, 19%, and 27%, respectively. The incidence of grade ≥ 3 adverse events (AEs) in arms A, B, and C was 65%, 55%, and 64%, respectively. Frequent AEs included hypertension (8%, 6%, and 5% in arms A, B, and C, respectively), peripheral edema (71%, 51%, and 22% in arms A, B, and C, respectively), and hypokalemia (21%, 15%, and 5% in arms A, B, and C, respectively). AMG 386 exhibited linear pharmacokinetic properties at the tested doses.

Conclusion

AMG 386 combined with weekly paclitaxel was tolerable, with a manageable and distinct toxicity profile. The data suggest evidence of antitumor activity and a dose-response effect, warranting further studies in ovarian cancer.

J Clin Oncol 30:362-371. © 2011 by American Society of Clinical Oncology

INTRODUCTION

Angiogenesis has been implicated in the development and progression of ovarian cancer.^{1,2} Angiopoietin-1 (Ang1) and angiopoietin-2 (Ang2) are ligands of the Tie2 receptor, which is expressed on vascular endothelial cells. The angiopoietin and vascular endothelial growth factor (VEGF) axes are separate pathways that play distinct roles in tumor angiogenesis.³ Evidence suggests that both are involved in pathologic angiogenesis associated with ovarian cancer.⁴⁻⁶ Treatment with

the VEGF inhibitor bevacizumab has been extensively studied and has demonstrated efficacy among patients with advanced ovarian cancer.⁷⁻¹⁴ Specifically, bevacizumab maintenance in combination with carboplatin/paclitaxel has shown improved progression-free survival (PFS) over carboplatin/paclitaxel plus placebo¹³ or carboplatin/paclitaxel alone in newly diagnosed patients.¹⁴ However, bevacizumab treatment may increase the risk of bowel perforation and other morbidities among patients with cancer¹⁵ and specifically among those with recurrent ovarian cancer.^{9,16-18}

AMG 386 is a first-in-class investigational peptide-Fc fusion protein that neutralizes the interaction between the Tie2 receptor and Ang1 and Ang2. In preclinical studies, Ang2 inhibition mediated antiangiogenic and antitumor activity in human tumor xenografts.¹⁹ Ang1 antagonism augmented Ang2 antagonism in curbing tumor growth, and dual Ang1/2 inhibition induced vessel regression, whereas selective Ang2 inhibition did not.²⁰ In phase I clinical trials in advanced solid cancers, AMG 386 treatment showed antitumor activity as monotherapy (first-in-human study with doses up to 30 mg/kg once weekly [QW])²¹ and in combination with chemotherapy (at the selected dose of 10 mg/kg),²² while presenting a distinct adverse event (AE) profile. In the first-in-human study, one patient with advanced refractory ovarian cancer who received AMG 386 at 30 mg/kg QW had a durable partial response per Response Evaluation Criteria in Solid Tumors (RECIST) that persisted until she withdrew from study at week 156.²¹ The aim of the present study was to evaluate the efficacy, tolerability, and pharmacokinetics of AMG 386 in combination with weekly paclitaxel in patients with recurrent ovarian cancer.

PATIENTS AND METHODS

Patients

Eligible women (≥ 18 years old) had histologically or cytologically documented epithelial ovarian (International Federation of Gynecology and Obstetrics stage II to IV), fallopian tube, or primary epithelial peritoneal cancer; recurrent measurable or nonmeasurable disease per RECIST (version 1.0)²³; or recurrent or nonmeasurable disease per CA-125 progression per Gynecologic Cancer Intergroup (GCI) criteria.²⁴ Other eligibility criteria were ≤ 3 previous anticancer therapies including ≥ 1 platinum-based regimen and Gynecologic Oncology Group performance status of ≤ 1 . Key exclusion criteria were a higher than average risk for bowel perforation (symptoms of partial or complete bowel obstruction, fistula or bowel perforation within 6 months, or requirement for total parenteral nutrition and continuous hydration); ongoing small bowel dysfunction; CNS metastases; prior malignancy (unless curatively treated); peripheral neuropathy grade ≥ 2 ; previous abdominal radiation therapy; inadequate cardiac, renal, hepatic, or hematologic function; bleeding diathesis within 14 days; and arterial or venous thrombosis within 12 months. Patients previously treated with angiopoietin inhibitors were excluded; prior VEGF/VEGF receptor inhibitor treatment was allowed with a sufficient washout period (60 days for long half-life molecules such as bevacizumab; 21 days for other agents). All patients provided written informed consent, and all study procedures were approved by the appropriate institutional ethics committees.

Study Design and Treatment

This randomized, double-blind, placebo-controlled, phase II study was an international collaboration among 38 centers. The primary end point was PFS using a composite definition (see Statistical Analysis). Secondary end points included overall survival, additional measures of efficacy, CA-125 response rate, change in CA-125 values, incidence of AEs, and pharmacokinetics of AMG 386.

All patients received intravenous paclitaxel 80 mg/m² QW (3 weeks on/1 week off) and were randomly assigned 1:1:1 using an automated voice response telephone system to also receive intravenous AMG 386 10 mg/kg (arm A), AMG 386 3 mg/kg (arm B) QW, or placebo QW (arm C) until progression, unacceptable toxicity, or withdrawal of consent. Random assignment was stratified by prior anti-VEGF therapy and disease progression on or within 6 months of the last chemotherapy. Treatment assignments were blinded to patients and all study site personnel until the primary analysis. Dose modifications for AMG 386 and placebo were not permitted. If grade ≥ 3 treatment-related toxicity occurred, AMG 386 or placebo was withheld until it resolved to grade ≤ 1 or baseline. Both treatments were discontinued if dosing was withheld for more than 28 days or withheld three times. Patients in arm C

were offered open-label AMG 386 10 mg/kg QW monotherapy after disease progression if they remained eligible.

Efficacy Assessments

Computed tomography or magnetic resonance imaging was performed every 8 ± 1 weeks during the study. Tumor responses per RECIST²³ were determined by the investigator. Assays for CA-125 were performed by both local and central laboratories. For patients with an assessment of CA-125 progression or response (per GCI criteria^{24,25}), a confirmatory measurement was required ≥ 28 days after the first criteria for CA-125 progression/response were met. Evaluable patients without confirmation of tumor and/or CA-125 response were considered nonresponders. Patients who discontinued treatment without evidence of progression and who had not withdrawn consent continued with radiographic and CA-125 assessments until progressive disease occurred or a new therapy was initiated.

Toxicity

All AEs were recorded and graded by the investigators according to the National Cancer Institute Common Terminology Criteria for Adverse Events (version 3.0).²⁶

Immunogenicity

Serum samples for assessment of human anti-AMG 386 binding and neutralizing antibodies (as previously described²¹) were collected before dose on day 1 of cycles 1 to 3 and every 16 weeks thereafter.

Pharmacokinetics

Serum samples were collected at the end of infusion at weeks 1 and 5 (maximum observed concentration [C_{max}]) and before infusions at weeks 1, 3, 5, and 9 and every 8 weeks thereafter (minimum observed concentration [C_{min}]). Serum AMG 386 concentrations were measured using an enzyme-linked immunosorbent assay; pharmacokinetic parameters were estimated using noncompartmental methods (WINNonlin version 5.1.1 on Citrix; Pharsight Corporation, Mountain View, CA).

Statistical Analysis

The planned enrollment of 150 patients ($n = 50$ patients per arm) was selected to generate treatment effect estimates. The primary end point was PFS; the primary statistical assessment was estimation of the PFS hazard ratio (HR). The primary analysis was planned when a total of 113 PFS events had occurred. HRs with two-sided 80% and/or 95% CIs were derived from a stratified Cox regression model for comparisons of PFS, PFS per RECIST, and time to progression (TTP) based on the intent-to-treat analysis set between arms A and B combined versus arm C (the same approach was used to estimate HRs for arms A and B separately). With a hypothesized HR for PFS of 0.75, an HR with a two-sided 80% CI with a maximum half-width of 0.22 (comparing arms A and B combined v arm C) was estimated.^{27,28} Pairwise comparisons between arms were performed using stratified log-rank tests. Although this was an estimation study, for 150 patients, it had approximately 57% power to detect the treatment effect (PFS: HR, 75%) of paclitaxel plus placebo ($n = 50$) versus paclitaxel plus AMG 386 ($n = 100$) at a 20% two-sided level of significance (chosen because of the randomized phase II design).

No adjustment for multiple testing was made when calculating CIs. Kaplan-Meier estimates of median survival times were calculated as previously described.²⁹ In prespecified descriptive analyses, dose-response effects for time-to-event end points among treatment groups were explored using stratified Tarone's tests.³⁰

Primary efficacy analyses included the intent-to-treat analysis set (data collected from patients after they had started to receive open-label AMG 386 were excluded from all efficacy analyses except overall survival). Safety analyses included data from the double-blind phase for all patients who received ≥ 1 dose of AMG 386 or placebo. PFS as the primary end point was defined as time from random assignment to disease progression per RECIST, clinical progression (per investigator) or CA-125 progression (per GCI criteria), or death. PFS defined as time from random assignment to disease progression per RECIST or death from any cause (PFS per RECIST) was a secondary end point. TTP was defined as time from random assignment to disease progression per RECIST, clinical progression, or CA-125 progression.

Table 1. Baseline Demographics and Clinical Characteristics

Demographic or Clinical Characteristic	Arm A (AMG 386 10 mg/kg QW + paclitaxel; n = 53)		Arm B (AMG 386 3 mg/kg QW + paclitaxel; n = 53)		Arm C (placebo + paclitaxel; n = 55)	
	No. of Patients	%	No. of Patients	%	No. of Patients	%
Race/ethnicity						
White	41	77	49	92	44	80
Asian	6	11	2	4	7	13
Black	3	6	1	2	0	0
Hispanic	2	4	1	2	2	4
Other	1	2	0	0	2	4
Age, years						
Median	59		60		62	
Range	27-80		28-85		38-83	
GOG performance status						
0	26	49	32	60	30	55
1	25	47	21	40	25	45
2-3*	2	4	0	0	0	0
Primary tumor type						
Ovarian cancer	46	87	47	89	44	80
Primary peritoneal cancer	7	13	6	11	8	15
Fallopian tube cancer	0	0	0	0	3	5
Tumor histology						
Serous	30	57	23	43	34	62
Endometrioid	7	13	6	11	3	5
Clear cell	1	2	0	0	2	4
Mucinous	0	0	1	2	1	2
Unclassified	12	23	21	40	13	24
Unavailable	3	6	2	4	2	4
FIGO disease stage at screening						
I*	1	2	0	0	2	4
II	1	2	1	2	1	2
III	22	42	29	55	25	45
IV	13	25	11	21	17	31
Unknown/unavailable	16	30	12	23	10	16
No. of prior anticancer therapies						
1	21	40	23	43	17	31
2	19	37	17	32	26	47
3	12	23	13	25	12	22
No. of lines of prior platinum therapy						
1	30	58	30	57	26	47
2	18	35	15	28	22	40
3	4	8	8	15	7	13
Platinum sensitivity status†						
Refractory	5	9	3	6	4	7
Resistant (PFI < 6 months)	20	38	23	43	20	36
Sensitive						
PFI 6 to 12 months	12	23	21	40	20	36
PFI > 12 months	15	28	6	11	10	18
Unavailable	1	2	0	0	1	2
Stratification factors						
Prior anti-VEGF/VEGFR therapy						
Yes	3	6	2	4	3	5
No	50	94	51	96	52	95
Disease progression on or within 6 months of the last chemotherapy regimen						
Yes	29	55	29	55	29	53
No	24	45	24	45	26	47

(continued in next column)

Table 1. Baseline Demographics and Clinical Characteristics (continued)

Demographic or Clinical Characteristic	Arm A (AMG 386 10 mg/kg QW + paclitaxel; n = 53)		Arm B (AMG 386 3 mg/kg QW + paclitaxel; n = 53)		Arm C (placebo + paclitaxel; n = 55)	
	No. of Patients	%	No. of Patients	%	No. of Patients	%
No. of AMG 386 or placebo infusions						
Median	24		20		13	
Range	1-106		1-55		2-83	
Follow-up time, weeks‡						
Median	66		65		64	
Range	40-120		40-112		40-110	

Abbreviations: FIGO, International Federation of Gynecology and Obstetrics; GOG, Gynecologic Oncology Group; PFI, platinum-free interval; QW, once weekly; VEGF, vascular endothelial growth factor; VEGFR, vascular endothelial growth factor receptor.

*Enrollment of patients with GOG performance status ≥ 2 or FIGO stage I disease was considered a protocol violation.

†The PFI was calculated as the time from the last dose of the last platinum-containing regimen until disease progression. Refractory is defined as patients who experienced progression during the first-line platinum-containing chemotherapy regimen; resistant is defined as patients who did not experience progression during the first-line therapy but who experienced disease progression during the last (second or later line) platinum-containing chemotherapy regimen or who experienced progression within 6 months of the last dose of the last (second or later line) platinum-containing regimen.

‡From date of random assignment to the date of data cutoff for the analysis.

RESULTS

Patients

One hundred sixty-one patients were randomly assigned between July 2007 and January 2009 (arm A, n = 53; arm B, n = 53; arm C, n = 55). One patient in arm A did not receive treatment because of grade 2 asthenia that occurred within 6 days of random assignment; all other randomly assigned patients received ≥ 1 dose. Baseline demographics and clinical characteristics were generally balanced across treatment arms (Table 1). Five percent of all patients had received prior anti-VEGF therapy, and 54% had disease progression on or within 6 months of the last chemotherapy. Median CA-125 levels at baseline in arms A, B, and C were 273 U/mL (range, 4 to 11,696 U/mL), 216 U/mL (range, 4 to 5,835 U/mL), and 157 U/mL (range, 4 to 6,000 U/mL), respectively. At the time of this primary analysis, 16 patients continued to receive AMG 386 or placebo (arms A, B, and C: n = 7, n = 5, and n = 4, respectively; Fig 1). The most common reasons for treatment discontinuation were disease progression (arms A, B, and C: n = 31, n = 25, and n = 37, respectively) and patient's decision to stop treatment/withdrawal of consent (arms A, B, and C: n = 5, n = 11, and n = 8, respectively). Ten patients in arm A, 13 in arm B, and 14 in arm C did not have radiographic or CA-125 progression. Seventeen patients in the placebo arm received AMG 386 in the open-label cross-over portion of the study; three patients have been continuing AMG 386 monotherapy for 2 to 10 months.

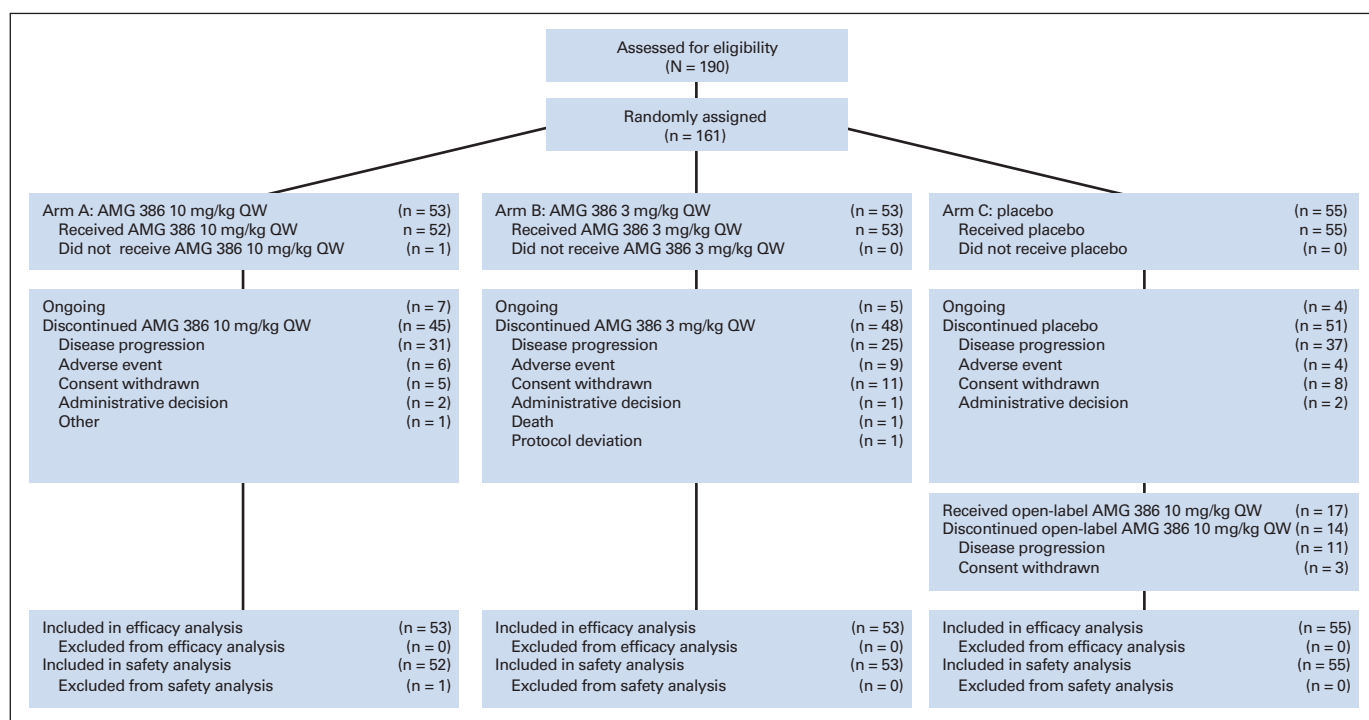


Fig 1. Flow diagram for study: disposition of patients. QW, once weekly.

PFS

At the time of this analysis, 43 patients in arm A, 40 in arm B, and 41 in arm C had experienced a PFS event defined as disease progression per RECIST, clinical progression, CA-125 progression, or death. Median PFS times in arms A and B were 7.2 and 5.7 months, respectively, compared with 4.6 months in arm C (Table 2; Fig 2). In a Cox model analysis of PFS adjusted for the stratification factors, the HR for arms A and B combined versus arm C was 0.76 (95% CI, 0.52 to 1.12; $P = .165$). Tarone's test and additional analyses exploring dose-exposure suggest a dose-response effect for PFS across the three arms ($P = .037$; Table 2). A subgroup analysis of PFS suggested that the magnitude of the treatment effect on PFS was greatest among patients who had received one prior therapy and, in arm A, among patients who were platinum refractory (Fig 3). HRs for TTP and PFS per RECIST showed a similar direction relative to placebo as seen with the primary end point results (Table 2). Of the randomly assigned patients, only eight had received prior anti-VEGF therapy. A sensitivity analysis that collapsed the prior anti-VEGF strata provided results consistent with the primary analysis (data not shown). At the time of this analysis, overall survival data were not mature. In an updated analysis (conducted 8 months after the primary analysis), median overall survival times in arms A, B, and C were 22.5, 20.4, and 20.9 months, respectively (Table 2).

Objective Response Rate and CA-125 Response

Most patients had measurable disease at baseline. The confirmed objective response rate was 37% in arm A, 19% in arm B, and 27% in arm C (Table 2). Duration of response and time to response were similar across treatments. Reductions in the target lesion dimensions from baseline measure according to RECIST occurred in 84%, 79%, and 64% of patients in arms A, B, and C, respectively (Fig 4A).

One hundred nineteen patients were evaluable for CA-125 response. The proportions of patients with a confirmed CA-125 response in arms A, B, and C were 71%, 58%, and 28%, respectively (Table 2). All evaluable patients in arm A had a decrease in serum CA-125 from baseline compared with 84% of patients in arm B and 63% of patients in arm C (Fig 4B).

Toxicity

All patients except two in the placebo arm experienced one or more AE (Table 3). The most frequently occurring toxicities were peripheral edema, fatigue, nausea, alopecia, and diarrhea. Although some AEs were more frequent in arm A than arms B or C (eg, peripheral edema), there was no clear trend in dose-related toxicity. Infusion reactions were rare and were only seen in one patient in arm A (grade 3) and in two patients who received placebo (grade 1). More patients in arms A (11%) and B (17%) had AEs leading to AMG 386/placebo discontinuation, compared with arm C (7%).

The incidence of grade 3 AEs was 54%, 34%, and 53% in arms A, B, and C, respectively; the incidence of grade 4 AEs was 10%, 15%, and 2%, respectively. Serious AEs occurred in 29%, 30%, and 33% of patients in arms A, B, and C, respectively. Grade ≥ 3 AEs occurring more frequently in arms A or B were hypokalemia, peripheral neuropathy, dyspnea (Table 3), anorexia (2%, 6%, and 0% in arms A, B, and C, respectively), and neutropenia (8%, 9%, and 4% in arms A, B, and C, respectively). One patient in arm A and three patients in arm B had grade 5 AEs compared with five patients in arm C. The only AE responsible for more than one death was ovarian cancer (arm C, $n = 4$). None of the deaths in arms A and B were considered related to AMG 386.

Several AEs generally of interest were noted (Table 3). One patient in arm C had grade 4 peritonitis as a result of bowel perforation;

Table 2. AMG 386 Efficacy Analysis

End Point	Arm A (AMG 386 10 mg/kg QW + paclitaxel; n = 53)	Arm B (AMG 386 3 mg/kg QW + paclitaxel; n = 53)	Arm C (placebo + paclitaxel; n = 55)
Primary end point: PFS per RECIST, clinical progression, CA-125 progression, or death*			
Kaplan-Meier PFS time, months			
Median	7.2	5.7	4.6
95% CI	5.3 to 8.1	4.6 to 8.0	1.9 to 6.7
Stratified Cox model			
PFS v arm C			
HR	0.76	0.75	
95% CI	0.49 to 1.18	0.48 to 1.17	
P	.225	.207	
PFS, arms A + B v arm C			
HR	0.76		
95% CI	0.52 to 1.12		
80% CI	0.59 to 0.98		
P	.165		
Dose-response test			
Tarone's test, P		.037	
Secondary end points			
PFS per RECIST or death†			
Kaplan-Meier PFS time, months			
Median	7.3	6.8	5.6
95% CI	5.5 to 8.9	5.3 to 8.7	3.6 to 7.6
Stratified Cox model			
PFS v arm C			
HR	0.81	0.75	
95% CI	0.51 to 1.30	0.49 to 1.21	
P	.384	.237	
PFS, arms A + B v arm C			
HR	0.80		
95% CI	0.53 to 1.20		
P	.275		
OS‡			
Deaths			
No.	22	28	28
%	42	53	51
Kaplan-Meier OS time, months			
Median	22.5	20.4	20.9
95% CI	20.4 to NE	17.7 to 28.5	11.3 to 24.2
Stratified Cox model			
OS v arm C			
HR	0.60	0.77	
95% CI	0.34 to 1.06	0.45 to 1.31	
P	.081	.330	
TTP per RECIST, clinical progression (per investigator), or CA-125 progression (per GCIG criteria)			
Kaplan-Meier TTP, months			
Median	7.3	7.4	4.8
95% CI	5.7 to 8.4	4.6 to 8.4	1.9 to 7.4
Stratified Cox model			
TTP v arm C			
HR	0.75	0.73	
95% CI	0.48 to 1.19	0.45 to 1.19	
P	.223	.207	
TTP arms A + B v arm C			
HR	0.75		
95% CI	0.50 to 1.12		
P	.155		

(continued in next column)

Table 2. AMG 386 Efficacy Analysis (continued)

End Point	Arm A (AMG 386 10 mg/kg QW + paclitaxel; n = 53)	Arm B (AMG 386 3 mg/kg QW + paclitaxel; n = 53)	Arm C (placebo + paclitaxel; n = 55)
Tumor response (per RECIST)			
Patients with measurable disease at baseline			
No.	46	47	52
%	87	89	95
Best response assessment			
No.	17	9	14
%	37	19	27
Complete response			
No.	2	1	0
%	4	2	0
Partial response			
No.	15	8	14
%	33	17	27
Stable disease			
No.	20	22	18
%	43	47	35
Progressive disease			
No.	6	8	13
%	13	17	25
Unevaluable§			
No.	0	0	1
%	0	0	2
Not done			
No.	3	8	6
%	7	17	12
Confirmed objective response rate, %			
95% CI	23 to 52	9 to 33	16 to 41
Time to response, months			
Mean	2.9	3.9	2.9
95% CI	2.2 to 3.7	2.8 to 5.1	2.3 to 3.6
Duration of response, months			
Median	5.4	5.7	7.1
95% CI	3.7 to 6.9	4.7 to NE	3.7 to NE
Best CA-125 response			
Patients evaluable			
No.	41	38	40
%	77	72	73
Confirmed CA-125 response			
No.	29	22	11
%	71	58	28

Abbreviations: CR, complete response; GCIG, Gynecologic Cancer Intergroup; HR, hazard ratio; NE, not estimable; OS, overall survival; PFS, progression-free survival; PR, partial response; QW, once weekly; RECIST, Response Evaluation Criteria in Solid Tumors; TTP, time to progression.

*Defined as the time from random assignment to disease progression per RECIST, clinical progression (per investigator) or CA-125 progression (per GCIG criteria), or death. At the time of this analysis, 10 patients in arm A, 13 patients in arm B, and 14 patients in arm C were censored.

†Defined as the time from random assignment to disease progression per RECIST or death from any cause.

‡At the time of the primary analysis, OS data were not mature. Data are from an updated analysis conducted 8 months after the primary analysis.

§Includes patients with a response assessment of complete response, partial response, or stable disease before the scheduled first assessment of response without an additional assessment of response.

||Includes patients for whom imaging was not performed at the scheduled assessment of response.

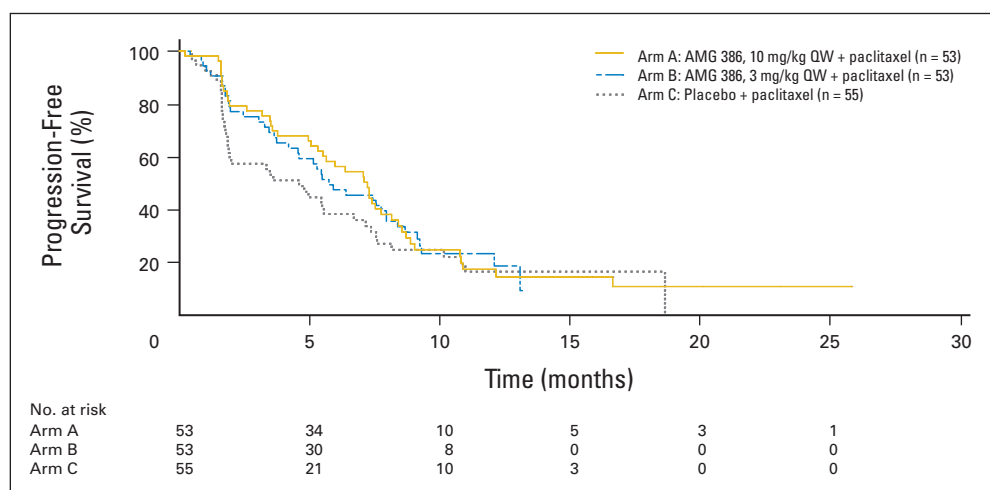


Fig 2. Progression-free survival (radiographic progression per Response Evaluation Criteria in Solid Tumors, clinical progression, CA-125 progression per Gynecologic Cancer Intergroup criteria, or death) for patients who received AMG 386 at 10 mg/kg once weekly (QW; arm A), AMG 386 at 3 mg/kg QW (arm B), or placebo (arm C).

no other perforations occurred. No patients experienced grade ≥ 3 hypertension, proteinuria, or impaired wound healing. The incidence of hemorrhagic events and arterial and venous thromboembolic events was similar across arms. One patient in arm B had a grade 5 pulmonary embolism.

In arms A and B, a total of two patients had pre-existing anti-AMG 386 antibodies, and three patients developed anti-AMG 386 antibodies during the study. No patients developed anti-AMG 386 neutralizing antibodies.

Pharmacokinetics

AMG 386 exhibited linear pharmacokinetic properties at doses of 3 and 10 mg/kg. Paclitaxel coadministration did not markedly affect AMG 386 exposure. Among patients who received AMG 386 10 mg/kg, mean C_{min} values at weeks 5 (19.2 $\mu\text{g/mL}$) and 9 (20.6 $\mu\text{g/mL}$), and every 8 weeks thereafter (27.4 $\mu\text{g/mL}$) and mean C_{max} values at week 5 (277 $\mu\text{g/mL}$) were consistent with those reported for the same dose in

the monotherapy study in advanced solid tumors (26.6 and 249 $\mu\text{g/mL}$, respectively).²¹ Similarly, at the 3 mg/kg dose, mean C_{min} values at weeks 5 (6.01 $\mu\text{g/mL}$) and 9 (6.93 $\mu\text{g/mL}$) and every 8 weeks thereafter (7.02 $\mu\text{g/mL}$) and mean C_{max} values at week 5 (81.9 $\mu\text{g/mL}$) were similar to those among patients who received the same dose as monotherapy (10.2 and 84.2 $\mu\text{g/mL}$, respectively).²¹

DISCUSSION

Although there was no statistically significant difference between treatment arms, estimates of median PFS obtained in this phase II study were encouraging among women with recurrent ovarian cancer who received weekly AMG 386 in combination with weekly paclitaxel, compared with placebo plus paclitaxel. Similar results were obtained for PFS per RECIST, TTP, objective response rate, and CA-125 response rate. CA-125 response is a widely used marker of response in ovarian cancer trials.³¹ Lower CA-125 levels during treatment have

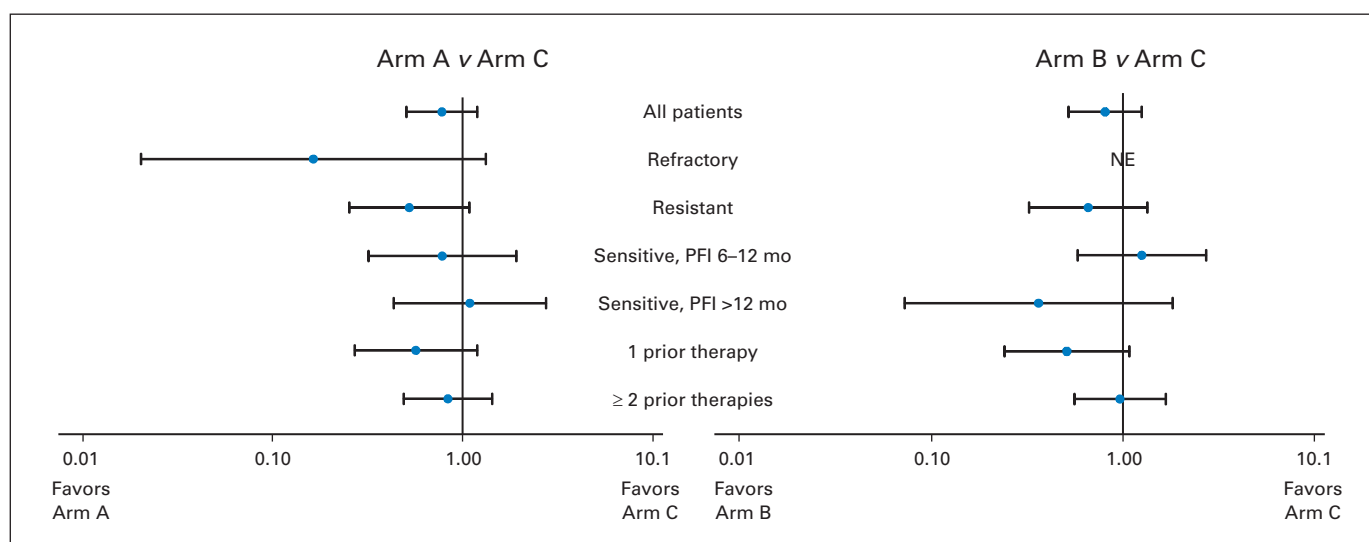


Fig 3. Comparison of progression-free survival (radiographic progression per Response Evaluation Criteria in Solid Tumors, clinical progression, CA-125 progression per Gynecologic Cancer Intergroup criteria, or death) among patient subgroups who received AMG 386 at 10 mg/kg once weekly (QW; arm A) and AMG 386 at 3 mg/kg QW (arm B) versus placebo (arm C). Data are hazard ratios $\pm$ 95% CIs. NE, not estimable; PFI, platinum-free interval.

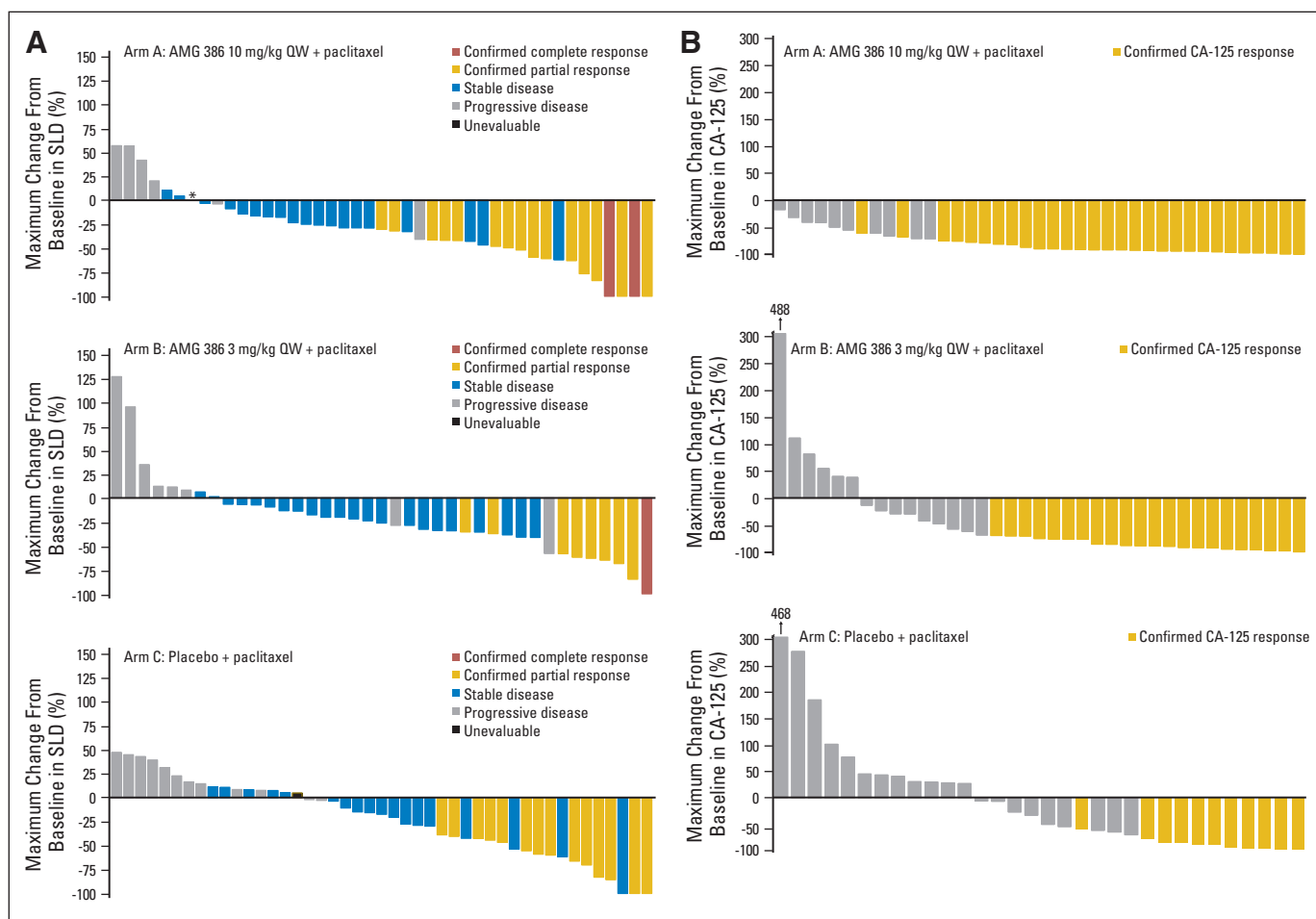


Fig 4. Best result change from baseline in (A) sum of longest diameters (SLD) for target lesions and (B) serum CA-125 for patients who received AMG 386 at 10 mg/kg once weekly (QW; arm A), AMG 386 at 3 mg/kg QW (arm B), or placebo (arm C). (*) Indicates patient with stable disease.

been associated with longer PFS^{32,33} and overall survival.³³ The consistent improvements in three clinical outcomes including PFS (regardless of definition) suggest meaningful clinical benefits associated with AMG 386 therapy in recurrent epithelial ovarian cancer. Subgroup analyses showed antitumor activity across patients in arm A with platinum-sensitive (PFI, 6 to 12 months), resistant, and refractory disease. Interestingly, among patients receiving AMG 386 10 mg/kg, those with refractory and resistant disease seemed to have derived greatest benefit. Treatment was generally well tolerated, with a distinct and manageable toxicity profile. The pharmacokinetic results, which confirm those from a phase IB study demonstrating that combining AMG 386 with chemotherapy is feasible,²² support repeated coadministration of paclitaxel and AMG 386.

The data also suggested a dose-response effect: AMG 386 10 mg/kg provided a 2.6-month increase in PFS over placebo, whereas AMG 386 3 mg/kg provided a 1.1-month increase. Results from a pharmacokinetic/pharmacodynamic analysis of the data suggest that maximum benefit in PFS may not have been reached with an AMG 386 dose of 10 mg/kg QW and that a dose of 15 mg/kg QW would provide greater exposure and yield further improvements in PFS.³⁴ In a phase I monotherapy study, AMG 386 was tolerable at doses up to 30 mg/kg without apparent changes in toxicity.²¹

The results of this study and others^{7-14,35} indicate that antiangiogenic agents have clinical activity in advanced ovarian cancer. Specifically, two large phase III studies have recently demonstrated that bevacizumab in combination with chemotherapy improves PFS among patients with previously untreated advanced ovarian cancer.^{13,14} However, some studies have suggested that bevacizumab treatment in recurrent ovarian cancer may result in an increased risk of bowel perforation.^{13,16-18} AMG 386 targets the Tie2-Ang1/2 signaling pathway. Therefore, it is not surprising that its toxicity profile is distinct from that of VEGF pathway inhibitors. The most frequently occurring AEs in the present study were peripheral edema, fatigue, nausea, and alopecia. There were no bowel perforations in either of the AMG 386 treatment arms. Two AEs emerged as likely being associated with AMG 386 treatment; these were peripheral edema and hypokalemia. Both occurred at higher overall incidence rates than with placebo, and there seemed to be more severe events (grade ≥ 3) of hypokalemia in patients who received AMG 386. The incidence rates of hypertension, thromboembolic events, proteinuria, and hemorrhage were similar across the active and placebo arms. Overall, the toxicity associated with AMG 386 treatment was manageable (the incidence of grade ≥ 3 AEs was similar in all arms) and was comparable to that previously reported for AMG 386 monotherapy²¹ and

Table 3. Incidence of Adverse Events

Adverse Event	Arm A (AMG 386 10 mg/kg QW + paclitaxel; n = 52)				Arm B (AMG 386 3 mg/kg QW + paclitaxel; n = 53)				Arm C (placebo + paclitaxel; n = 55)			
	All Grades		Grade ≥ 3		All Grades		Grade ≥ 3		All Grades		Grade ≥ 3	
	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Patients with any adverse event	52	100	34	65	53	100	29	55	53	96	35	64
Adverse events occurring in ≥ 25% of patients in 1 or more treatment arms												
Peripheral edema	37	71	2	4	27	51	3	6	12	22	2	4
Fatigue	34	65	1	2	37	70	3	6	26	47	2	4
Nausea	27	52	1	2	31	58	1	2	17	31	1	2
Alopecia	26	50	0	0	20	38	1	2	20	36	0	0
Diarrhea	22	42	1	2	19	36	2	4	15	27	0	0
Peripheral neuropathy	20	38	5	10	14	26	1	2	17	31	2	4
Constipation	20	38	1	2	12	23	0	0	15	27	0	0
Cough	18	35	0	0	16	30	0	0	9	16	1	2
Abdominal pain	17	33	1	2	17	32	2	4	18	33	3	5
Pain in extremity	15	29	0	0	15	28	0	0	8	15	0	0
Headache	15	29	0	0	13	25	0	0	7	13	0	0
Vomiting	14	27	2	4	14	26	3	6	9	16	2	4
Dizziness	14	27	0	0	6	11	0	0	10	18	0	0
Dyspnea	13	25	1	2	20	38	5	9*	12	22	2	4
Arthralgia	13	25	0	0	8	15	0	0	8	15	0	0
Urinary tract infection	13	25	0	0	6	11	1	2	7	13	0	0
Oropharyngeal pain	13	25	0	0	6	11	0	0	8	15	0	0
Back pain	12	23	2	4	13	25	0	0	10	18	0	0
Adverse events of interest												
GI perforation	0	0	0	0	0	0	0	0	1	2	1	2†
Hypertension	4	8	0	0	3	6	0	0	3	5	0	0
Arterial thromboembolic events	1	2	1	2	1	2	1	2‡	0	0	0	0
Venous thromboembolic events	4	8	3	6§	3	6	2	4	6	11	5	9¶
Hemorrhagic events	14	27	2	4	14	26	1	2	13	24	0	0
Impaired wound healing	1	2	0	0	1	2	0	0	0	0	0	0
Proteinuria	3	6	0	0	4	8	0	0	2	4	0	0
Infusion reactions	1	2	1	2	0	0	0	0	2	4	0	0
Hypokalemia	11	21	6	12#	8	15	6	11	3	5	2	4

NOTE. Adverse events are reported for all patients who received ≥ 1 dose of AMG 386 or placebo (safety analysis set). Unless otherwise specified, all events are of grade ≤ 3 in severity.

Abbreviation: QW, once weekly.

*Includes one patient with grade 5 dyspnea.

†Grade 4 peritonitis.

‡Grade 4 myocardial infarction.

§Includes two patients with a grade 4 pulmonary embolism.

||Includes one patient with a grade 5 pulmonary embolism and one with a grade 4 pulmonary embolism.

¶Includes one patient with a grade 4 pulmonary embolism.

#Includes two patients with grade 4 hypokalemia.

chemotherapy combination studies.²² No new toxicity signals were identified, and there were no dose-related trends with respect to AEs.

The key limitation of this phase II study is the relatively small number of patients enrolled. Although AMG 386 in combination with paclitaxel (arms A/B) showed clinically meaningful PFS results, the study was designed only to provide an estimation of the HRs for AMG 386 versus placebo. Larger studies will be required to provide a more definitive demonstration of the efficacy of AMG 386 in combination with chemotherapy in advanced recurrent ovarian cancer.

In summary, the results of this estimation study suggest that inhibiting angiogenesis via Tie2/angiopoietin pathway inhibition may offer an effective approach for the treatment of advanced recurrent ovarian cancer. Treatment with once-weekly AMG 386 plus paclitaxel may result in prolonged median PFS. The therapy was well tolerated with a distinct and manageable toxicity profile. The dose-response effect suggests that the maximum clinically effective dose may not have yet been reached. Once-weekly AMG 386 plus paclitaxel for the treatment of recurrent ovarian

cancer is being further investigated in a phase III study (TRINOVA-1 [Trial in Ovarian Cancer 1]; ClinicalTrials.gov, NCT01204749).

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Although all authors completed the disclosure declaration, the following author(s) indicated a financial or other interest that is relevant to the subject matter under consideration in this article. Certain relationships marked with a "U" are those for which no compensation was received; those relationships marked with a "C" were compensated. For a detailed description of the disclosure categories, or for more information about ASCO's conflict of interest policy, please refer to the Author Disclosure Declaration and the Disclosures of Potential Conflicts of Interest section in Information for Contributors.

Employment or Leadership Position: Daniel E. Stepan, Amgen (C); David M. Weinreich, Amgen (C); Marjan Tassoudji, Amgen (C); Yu-Nien Sun, Amgen (C) **Consultant or Advisory Role:** None **Stock Ownership:** Daniel E. Stepan,

Amgen; David M. Weinreich, Amgen; Marjan Tassoudji, Amgen; Yu-Nien Sun, Amgen **Honoraria:** Gary E. Richardson, Amgen (advisory board) **Research Funding:** Beth Y. Karlan, Amgen; Gary E. Richardson, Amgen – AMG 386 Study, Amgen – Denosumab Studies; Vincent L. Hansen, Amgen; John V. Brown III, Amgen; Charles A. Leath III, Amgen **Expert Testimony:** None **Other Remuneration:** None

AUTHOR CONTRIBUTIONS

Conception and design: Beth Y. Karlan, Daniel E. Stepan, David M. Weinreich, Marjan Tassoudji, Yu-Nien Sun

Provision of study materials or patients: Amit M. Oza, Diane M. Provencher, Setsuko K. Chambers, John V. Brown III, Allan Covens, Charles A. Leath III, Ignace B. Vergote
Collection and assembly of data: Beth Y. Karlan, Gary E. Richardson, Diane M. Provencher, Vincent L. Hansen, Martin Buck, Charles H. Pippitt Jr, John V. Brown III, Allan Covens, Raj V. Nagarkar, Margaret Davy, Hoa Nguyen, Yu-Nien Sun, Ignace B. Vergote
Data analysis and interpretation: Beth Y. Karlan, Amit M. Oza, Setsuko K. Chambers, Prafull Ghatage, John V. Brown III, Allan Covens, Charles A. Leath III, Daniel E. Stepan, David M. Weinreich, Marjan Tassoudji, Yu-Nien Sun, Ignace B. Vergote
Manuscript writing: All authors
Final approval of manuscript: All authors

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Affiliations

Beth Y. Karlan, Cedars-Sinai Medical Center, Los Angeles; John V. Brown III, Gynecologic Oncology Associates, Newport Beach; Daniel E. Stepan, David M. Weinreich, and Yu-Nien Sun, Amgen, Thousand Oaks; Marjan Tassoudji, Amgen, South San Francisco, CA; Vincent L. Hansen, Northern Utah Associates, Ogden, UT; Setsuko K. Chambers, Arizona Cancer Center, Tucson, AZ; Charles H. Pippitt Jr, Piedmont Hematology Oncology Associates, Winston-Salem, NC; Charles A. Leath III, Brooke Army Medical Center, Fort Sam Houston, TX; Hoa Nguyen, Sheridan Healthcare Corporation, Hollywood, FL; Amit M. Oza, Princess Margaret Hospital; Allan Covens, Toronto-Sunnybrook Regional Cancer Centre, Toronto, Ontario; Diane M. Provencher, Centre Hospitalier de l'Université de Montréal-Hôpital Notre-Dame, Montreal, Quebec; Prafull Ghatage, Tom Baker Cancer Centre, Calgary, Alberta, Canada; Gary E. Richardson, Cabrini Hospital, Melbourne, Victoria; Martin Buck, Sir Charles Gairdner Hospital, Perth, Western Australia; Margaret Davy, Royal Adelaide Hospital, Adelaide, South Australia, Australia; Raj V. Nagarkar, Curie Manavata Cancer Centre, Nashik, Maharashtra, India; and Ignace B. Vergote, University Hospital Leuven, Leuven, Belgium.



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